Inequalities and Limit theorems

(see Trivedi; Ross; Allen ..)

If distribution function (or equivalently pdf or pmf) is known, one can have complete characterization ie one can compute various moments, also probability of any event.

Question: Probability distribution – Not given

Information regarding first moment of a non-negative rv is given, can we obtain bounds on the probabilities?

We derive Markov inequality for non-negative rv. Examples of such rv are:

(i) Time to failure of a component

(ii) Response time in an interactive computer system

(iii) Number of items produced in a factory; etc.

Markov Inequality Let X be a non-negative rv such that $P[X<0]=0$. Further Expected value $E[X]$ is given. Then for each $t>0$

$$P[X \geq t] \leq \frac{E[X]}{t}.$$

Proof (a) We give the proof for the case when X is continuous with pdf f.

$$E[X] = \int_0^\infty x f(x)\,dx = \int_0^t x f(x)\,dx + \int_t^\infty x f(x)\,dx$$

$$E[X] \geq \int_t^\infty x f(x)\,dx \geq \int_t^\infty t f(x)\,dx = t\int_t^\infty f(x)\,dx$$

$$= t\,P(X \geq t).$$

or $P(X \geq t) \leq \dfrac{E[X]}{t}$

<u>(b)</u> We give proof <u>for discrete X with pmf</u>

$p_X(x_i) = P(X = x_i)$, $i = 1, 2, \ldots$

$$E[X] = \sum_{x_i} x_i\, p(x_i) = \sum_{x_i < t} x_i\, p(x_i) + \sum_{t \leq x_i} x_i\, p(x_i)$$

$$\geq \sum_{t \leq x_i} x_i\, p(x_i) \geq \sum_{t \leq x_i} t\, p(x_i) = t\,P[X \geq t]$$

Hence $P[X \geq t] \leq E[X]/t$, $t > 0$

<u>ANOTHER PROOF</u>

Fix $t > 0$ and define the rv Y

$$Y = \begin{cases} 0 & \text{if } X < t \\ t & X \geq t \end{cases}$$

Thus Y is a discrete rv with pmf

$p_Y(0) = P[X < t]$ and $p_Y(t) = P[X \geq t]$

We have $E[Y] = 0 \cdot p_Y(0) + t\, p_Y(t)$

$= t\,P[X \geq t]$

Note $X \geq Y$, we have $E[X] \geq E[Y]$

and hence $E[X] \geq t\,P[X \geq t]$

we have $P[X \geq t] \leq \dfrac{E[X]}{t}$

Example 1

Let X be a non-negative rv with $E[X] = 500$. Find the probability that X will be atleast 1000.

Soln.: By Markov's inequality

$$P[X \geq 1000] \leq \frac{E[X]}{1000} = \frac{500}{100} = \frac{1}{2}.$$

Example 2 Suppose an interactive computer system has the mean response time $E[T] = 0.5$ sec. Estimate the probability that the response time T will be 2 sec or more.

Soln By Markov's inequality

$$P[T \geq 2] \leq \frac{E[T]}{2} = \frac{0.5}{2} = \frac{1}{4}$$

Note Markov inequality gives rather crude estimates because we are using only the value of the first moment (see Allen)

For example

$$P[T \geq \tfrac{1}{2}] \leq \frac{0.5}{0.5} = 1$$

How to improve probability estimates?

— If (both) mean and variance are known. The better estimate can be provided by Chebyshev's inequality.

Chebyshev's Inequality

If X is a rv with mean μ and variance σ^2, then for any $k > 0$,

$$P(|X-\mu| \geq \frac{\sigma^2}{k^2}$$

Proof Note $(X-\mu)^2$ is non-negative rv

We can apply Markov's inequality

$$P((X-\mu)^2 \geq k^2) \leq \frac{E[(X-\mu)^2]}{k^2} = \frac{\sigma^2}{k^2}$$

or $P(|X-\mu| \geq k) \leq \frac{\sigma^2}{k^2}$

(Note $(X-\mu)^2 \geq k^2$ implies $|X-\mu| \geq k$)

Importance: These inequalities enable us to obtain estimates of probabilities or bounds when only the mean or both mean and variance are known

Example 3 In Example 1, we have $\mu = 500$. Further if $\sigma^2 = 100$, find $P(|X-500| \geq 100)$

Soln By Chebyshev's inequality

$$P(|X-500| \geq 100) \leq \frac{\sigma^2}{100^2} = \frac{100}{(100)^2} = \frac{1}{100}$$

Assignment (Session I) 17.3.2020

— p1 —

Q1. Use Markov's inequality to estimate

$$P[X \geq k E[X]] \; ; \; k > 0.$$

Q2 Let X denote the execution time of a job on a server and assume $X \sim Exp(\lambda)$,

(i) Determine

$$P\left(\left|X - \frac{1}{\lambda}\right| \geq \frac{1}{\lambda}\right).$$

(ii) If you have only information about the mean and variance of rv. i.e $E[X] = \frac{1}{\lambda}$ and $Var\, X = \frac{1}{\lambda^2}$; use Chebyshev's inequality to obtain estimate $P\left(\left|X - \frac{1}{\lambda}\right| \geq \frac{1}{\lambda}\right)$.

Comment on the result.

Q3. Derive alternate forms of Chebyshev's inequality, viz

(i) $P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$

(ii) $P(|X - \mu| \leq k\sigma) \geq 1 - \frac{1}{k^2}$.

Q4. Apply chebyshev's inequality to binomial distribution with $\mu = np$ and $\sigma^2 = npq$.

Show that

$$P\left(\left|\frac{X}{n} - p\right| < \epsilon\right) \geq 1 - \frac{p(1-p)}{n\epsilon^2}$$

For any given $\epsilon > 0$, interpret the result as $n \to \infty$.

Q1.

$$P[X \geq k\,E(X)] \leq \frac{E(X)}{k\,E(X)} = \frac{1}{k}, \quad k>0$$

Q2.

(i) $P(|X-\frac{1}{\lambda}| \geq t) = P(0 \leq X \leq \frac{1}{\lambda}-t$ or $\frac{1}{\lambda}+t \leq X < \infty)$

$$= F_X(\tfrac{1}{\lambda}-t) + [1-F(\tfrac{1}{\lambda}+t)]$$

$$= 1-e^{-\lambda(\frac{1}{\lambda}-t)} + e^{-\lambda(\frac{1}{\lambda}+t)}$$

$$= 1-e^{+\lambda t-1} + e^{-\lambda t-1}$$

Note if $X \sim Exp(\lambda)$, then $F_X(x) = 1-e^{-\lambda x}$, $x>0$

Thus $P(|X-\frac{1}{\lambda}| \geq \frac{1}{\lambda}) = e^{-2}$

(ii) using chebyshev's inequality

$$P(|X-\tfrac{1}{\lambda}| \geq \tfrac{1}{\lambda}) \leq \frac{\sigma^2}{\frac{1}{\lambda^2}},$$

$$= \frac{1/\lambda^2}{1/\lambda^2} = 1$$

i.e. $P(|X-\frac{1}{\lambda}| \geq \frac{1}{\lambda}) \leq 1.$

This inequality does not yield any information.

Solution

Q4.

$$P\left(|X-np| < k\sqrt{np(1-p)}\right) \geq 1-\frac{1}{k^2}$$

or $$P\left(\left|\frac{X}{n}-p\right| < k\sqrt{\frac{p(1-p)}{n}}\right) \geq 1-\frac{1}{k^2}$$

Substituting $\epsilon = k\sqrt{p(1-p)/n}$, we get

$$P\left(\left|\frac{X}{n}-p\right| < \epsilon\right) \geq 1-\frac{p(1-p)}{n\epsilon^2}$$

For given value of $\epsilon > 0$,

$$\lim_{n\to\infty} P\left(\left|\frac{X}{n}-p\right| < \epsilon\right) = 1.$$

$\frac{X}{n}$ represents observed proportion of successes

$P\left(\frac{X}{n}\right.$ differs from p by less than any number ϵ)

approaches 1 as $n \to \infty$.